BUSINESS MATHEMATICS

MAX MARKS: 100 **TIME: 3 HOURS**

80 (THEORY) +20 (INTERNAL ASSESSMENT)

Unit 1st: Sets, Relations and Functions — 10 Marks

Sets and their representation, various types of sets ,compliment of a set . Algebra of sets (Union, intersection and difference of sets). Demorgan's laws, Cartesian product of sets .

Relations: Various types of relations, Equivalence relation simple examples

Definition of a function and its various types (Into, onto, one-one, many-one, polynomial function, rational, modulus, constant , signum, greatest integer function, composite function) .

Unit 2nd: Sequences and Series — 10 Marks

Geometric progression, general term sum to n terms, and sum to infinity of a geometric series . Geometric and arithmetic means , Evaluation of Σn , Σn^2 , Σn^3

Unit 3rd: Trigonometry — 10 Marks

Trigonometric ratios of allied angles (without proof). sum, difference formulae and their applications . solution of trigonometric equations .

Unit 4th : Permutations and combinations — 10 Marks

Factorial notation, fundamental principle of counting. Meaning of P(n,r) and C (n,r) and their relations with simple applications.

Unit 5th: Binomial theorem — 08 Marks

Binomial theorem for any index. General term, middle term/s of a Binomial Expansion. Applications of binomial expansion.

Unit 6th: Statistics — 13 Marks

Measures of dispersion, Mean Deviation from mean and median.

Standard deviation and variance of a grouped and ungrouped data.

Quartile deviation.

Unit 7th: Probability — 11 Marks

Random experiment and sample space (set representation). Events and their occurrence. various types of events. Mutually exclusive and Exhaustive events . Axiomatic probability with applications.

Unit 8th: Linear Inequations — 08 Marks

Algebraic solution of an inequation in one variable and the representation on a number line. Graphical solution of linear inequations in two variables.

Books Suggested:

1. A Textbook of Mathematics for Class XI published by NCERT, New Delhi

Scheme of Assessment (Theory)

Question paper contains four sections namely

1. **Section –A** contains 10 (Objective Type Question/Multiple Choice Questions) of 1 mark each. 1x10= 10 marks
2. **Section- B** contains 10 Very Short Answer Type Questions of 2 marks each. 2 x 10 = 20 marks
3. **Section – C** contains 8 Short Answer type questions of 4 marks each. 4x 8 = 32 marks
4. **Section - D** contains 3 Long Answer type questions of 6 marks each. (With Internal Choice) 6 x 3 = 18 marks

Note: The paper setter shall incorporate Higher Order Thinking Skills (HOTS) questions apart from knowledge, memory and application based questions.

INTERNAL ASSESSMENT	20 MARKS
Periodic Tests (Best 2 out of 3 tests conducted)	10 Marks
Mathematics Activities	10 Marks